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Sensing and throughput analysis of a MU-MIMO based cognitive radio scheme for the Internet of Things

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ABSTRACT

State-of-the-art energy detection (ED) based spectrum sensing requires perfect knowledge of noise power and is vulnerable to noise uncertainty. An eigenvalue-based spectrum sensing approach performs well in such an uncertain environment, but does not mitigate the spectrum scarcity problem, which evolves with the future Internet of Things (IoT) rollout. In this paper, we propose a multi-user multiple-input and multiple-output (MU-MIMO) based cognitive radio scheme for the Internet of Things (CR-IoT) with weighted-eigenvalue detection (WEVD) for the analysis of sensing, system throughput, energy efficiency and expected lifetime. In this scheme, each CR-IoT user is being equipped with MIMO antennas; we calculate the WEVD ratio, which is defined as the ratio between the difference of the maximum eigenvalue and minimum eigenvalue to the sum of the maximum eigenvalue and minimum eigenvalue. This mitigates against the spectrum scarcity problem, enhances system throughput, improves energy efficiency, prolongs expected lifetime and lowers error probability. Simulation results confirm the effectiveness of the proposed scheme; here the WEVD technique demonstrates a better detection gain and enhanced system throughput in comparison to the conventional scheme with eigenvalue based detection (EVD) and ED techniques in a noise uncertainty environment (i.e. SNR < -28). Furthermore, the proposed scheme has a lower energy consumption, prolonged expected lifetime and achieves a low error probability when compared with other schemes like the conventional single-input and single-output (SISO) based CR-IoT scheme with EVD and ED spectrum sensing.

1. Introduction

The Internet of things (IoT) is a new paradigm of (wireless) machine-to-machine (M2M) communication, which enables machines or devices to interact with each other without human intervention. IoT constitutes networked devices that gather and exchange information of different objects using different communication technologies [1]. IoT applications are used for information exchange and analysis [2]. The scarcity of the available radio spectrum used by an ever increasing number of IoT devices is a big challenge, as the spectrum allocation mechanisms used in IoT communications are fixed and therefore underutilized [3]. Moreover, ease of access, bandwidth allocation and utilization and availability of spectrum are major concerns to future IoT networks, particularly with the emergence of data-intense IoT application areas such as health-care, environmental monitoring and smart cities [4–7]. Therefore the major challenges for future IoT communication are efficient spectrum utilization, enhanced system throughput, and lower energy consumption, hereby considering the fact that most IoT devices will not have the hardware capabilities or energy resources to perform sophisticated spectrum sensing methods over extended periods of time [8]. New spectrum sensing techniques will change the paradigm of spectrum access by increasing utilization of the licensed spectrum.

Cognitive radio (CR) has been identified as a revolutionary technology to meet the future IoT spectrum demand [6,7], as it dynamically or statically identifies the vacant channels of a spectrum band used by primary users (PUs), and temporarily/dynamically allocate it to IoT devices. CR-enabled IoT devices form a CR-IoT network that coexists with a PU network [9,10]. CR-IoT devices must not deny the PU network access when needed, and only opportunistically access the unused licensed spectrum.

CR technologies can be categorized into the following two approaches [11]: (i) the overlay approach, where the CR-IoT users identify an unused portion of the licensed bands through a spectrum sensing process and opportunistically access the available spectrum for data transmission; (ii) the underlay approach, in which CR-IoT communication with very low transmit power coexists with that of the PU in the same channel, so that the CR-IoT devices do not interfere with the communication of the PU.

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Spectrum sensing is a preliminary and fundamental function of CR systems and plays a vital role in a CR-IoT network that uses the overlay approach [12]. There are a number of spectrum sensing techniques: cyclostationary detection, matched filter detection, eigenvalue based detection and energy detection (ED). Cyclostationary detection offers good detection performance when the PU cyclic frequencies are known [13,14]. However, it requires a longer time slot for the sensing phase in the frame structure compared to matched filter detection [15], which in turn also requires complete knowledge of the PU's signaling features, such as bandwidth, operating frequency, modulation type and order, pulse shaping and packet format. Eigenvalue based detection is one of the most recent and promising techniques; here the test statistics is computed from the eigenvalues of the received signal sample covariance matrix [16]. Moreover, this method does not require prior knowledge of sampled signals. Finally, energy detection is a popular and widely used method for detecting an unknown signal. It has low complexity because it can be directly implemented without any prior information of the primary signal [17]. However, the performance of ED is very poor in a noise uncertainty environment [18]. It requires an expensive analog-to-digital converter that is able to oversample the received signal at more than 8 samples/symbol, as well as knowledge of the PU pulse shaping filter, which is impracticable in some cases [19].

Moreover, a single CR-IoT user cannot identify the PU signal accurately due to the hidden terminal problem, which is caused by multi-path fading and the shadowing effect. In order to diminish the effects of this problem, a cooperative spectrum sensing (CSS)/multi-user (MU) based CR-IoT network using a centralized controller or fusion center (FC) enhances detection gain compared to a single CR-IoT user. However, CSS/MU requires a reliable reporting channel to exchange local decisions between CR-IoT users.

With CSS each CR-IoT device performs local sensing independently and then forwards the sensing results to the corresponding FC through a (noise free or non-noise free) reporting channel. The FC collects all sensing results and makes a final global decision based on fusion rules. These fusion rules can be classified as soft decision fusion (SDF) or hard decision fusion (HDF) [20,21]. In SDF, each CR-IoT user forwards the entire sensing result to the FC, which makes a decision based on square law combining, maximal ratio combining and selection combining. While it provides better performance than HDF, it requires large bandwidth for the control channel. In HDF, each CR-IoT user makes a one-bit decision regarding the existence of the PU (using on AND, OR and M logic) and sends this decision to the FC, therefore only needing limited bandwidth. However, the sensing performance of HDF is lower than of SDF.

Modern multiple-input and multiple-output (MIMO) antennas herald a new dimension in spectrum sensing, as they provide high spectral efficiency, high data rates and extreme spatial multiplexing [22–27]. The concepts of MIMO has drawn a lot of interest in the CR research, as it can mitigate fading and shadowing effects. With the emergence of MIMO antennas, the multipath issue is effectively converted into a benefit for communication systems. For these reasons, in the proposed MU-MIMO based CR-IoT scheme, each CR-IoT device is being equipped with MIMO antennas and is called a MIMO based CR-IoT device. The proposed MU-MIMO based CR-IoT scheme can provide outstanding performance for detecting the PU signal compared to single-input and single-output (SISO).

Motivated by these observations, we propose a MU-MIMO based CR-IoT scheme using the WEVD technique for the analysis of sensing gain, system throughput, energy efficiency, expected lifetime and error probability. Moreover, we contend that our proposed approach is scalable in large CR-IoT networks. To motivate our assertion, we consider the scenario of a mobile phone cell with different frequency bands that are used to accommodate large numbers of PUs. Here each frequency band contains a number of CR-IoT users who must vacate the frequency band once the PU starts to broadcast on the same frequency. A FC for each frequency band will establish the presence or absence of the PU, scaling linearly as a function of the number of frequency bands, CR-IoT users and PUs [28,29].

In this paper, we have achieved the following major contributions:

- We propose a MU-MIMO based CR-IoT scheme that uses the WEVD technique for the analysis of sensing gain, system throughput, energy consumption and expected lifetime in a noise uncertainty environment; here the weighted-eigenvalue ratio is utilized by each CR-IoT user.
- We assess sensing performance gain, maximization of system throughput, energy efficiency and expected lifetime of the conventional scheme.
- We propose a MU-MIMO based CR-IoT scheme, in which each CR-IoT user can acquire more samples in a noise uncertainty environment, compared to a single-antenna based scheme.
- For the proposed scheme with the WEVD technique we empirically analyze the sensing gain at the FC using the HDF rule, i.e. the "OR-rule" and "M-rule".
- Based on the sensing gain, the system throughput of all PU and the CR-IoT network is analyzed for (i) the proposed scheme with the WEVD technique, (ii) the conventional scheme with the EVD technique and (iii) the conventional scheme with the ED technique.
- Moreover, based on the sensing gain and the system throughput, the energy efficiency and the expected lifetime are analyzed for (i) the proposed scheme with the WEVD technique, (ii) the conventional scheme with the EVD technique and (iii) the conventional scheme with the ED technique.
- Finally, simulation results show that the proposed scheme with the WEVD technique achieves both better detection gain and an enhanced system throughput compared to the conventional scheme with the EVD and ED techniques in a noise uncertainty environment. In addition, it is shown that the proposed scheme with the WEVD technique obtains a more efficient energy consumption, a prolonged expected lifetime and a lower error probability compared to the conventional scheme with the EVD and ED techniques.

The rest of the paper is organized as follow: The general motivation and the background of this paper is summarized in Section 2. In Section 3 we describe the proposed MU-MIMO based CR-IoT network system model, including the characterization of (i) the PU and CR-IoT user link and (ii) the CR-IoT user and fusion center link; here we explain the conventional scheme with the ED technique, the conventional scheme with the EVD technique and the proposed scheme with the WEVD technique. In Section 4, we explain the throughput and the energy consumption constraint, while providing a time complexity analysis of the proposed scheme. The simulation parameters of the proposed scheme and comparisons are given in Section 5. We compare our proposed scheme with the WEVD technique with the two other existing conventional schemes, and demonstrate better detection gain, enhanced system throughput, more efficient energy consumption, prolonged lifetime and a lower error probability in terms of detection performance. Finally, our conclusion and future work of the paper are addressed in Section 6.

Additionally, parameters used in this paper are summarized in Table 1 as follows:

2. Related work

The authors already presented an analysis of non-CSS in a cognitive radio network (CRN) [30,31]. However, the detection performance was not analyzed under the hidden terminal problem. An analysis of CSS in a CRN is presented in [32,33]. The authors analyzed only the cooperative detection performance under non-fading. However, they were not analyzed the cooperative detection performance under fading, shadowing, uncertainty and the hidden terminal problem. An efficient

Table 1
Main parameters.

Parameters	Meaning
K	The number of antennas
M	The number of CR-IoT users in the proposed scheme.
τ_s	Duration of sensing time slot
τ_r	Duration of reporting time slot
B	The channel bandwidth
$[.]^T$	Denote a transpose operation
$E[.]$	Denote a expectation operation
p_e	The global error probability at the FC
N_x^c	The total number of signal samples in the conventional scheme with the ED technique
N_x^p	The total number of signal samples in the proposed scheme with the WEVD technique
$Cov_y(N_x^c)$	Covariance matrix for the received signal vector in the conventional scheme with the EVD technique
$\overline{Cov_y}(N_x^p)$	Covariance matrix for the received signal vector in the proposed scheme with the WEVD technique
r_{mtm}^c	The maximum-to-minimum ratio in the conventional scheme with the EVD technique
r_{mtm}^p	The maximum-to-minimum ratio in the proposed scheme with the WEVD technique
λ_{EVD}^c	The decision threshold in the conventional scheme with the EVD technique
λ_{WEVD}^p	The decision threshold in the proposed scheme with the WEVD technique
$p_{i,k}^f$	The local probability of false alarm in the conventional scheme with the ED technique of the ith CR-IoT user with k antennas
$p_{i,k}^d$	The local probability of detection in the conventional scheme with the ED technique of the ith CR-IoT user with k antennas
gd_{ED}^f	The global probability of false alarm in the conventional scheme with the ED technique
gd_{ED}^d	The global probability of detection in the conventional scheme with the ED technique
$gd_{f\|d}$	The global probability of false alarm or probability of detection in the conventional scheme with the EVD technique
$\overline{gd}_{f\|d}$	The global probability of false alarm or probability of detection in the proposed scheme with the WEVD technique

transmission mode selection based on reinforcement learning for cooperative CRNs is studied in [33]. It is shown that the spectrum sensing accuracy is compromised because of fading, shadowing, uncertainty and the hidden terminal problem [34–36]. The advantages of CSS can be limited due to noisy reporting channels which are studied in [37–40]. In a CRN with a large number of cognitive radio users (CRUs) CSS requires many reports from CRUs to the FC through the control channel, which can result in overhead traffic of the control channel. Some methods have been proposed to solve this problem with CSS such as sequential CSS [41,42]. However, the sensing performance of this scheme may be decreased due to fading effects of reporting channels in real environments. The authors previously proposed large-scale MU-MIMO approach for cognitive radio sensor networks [43]. However, the sensing and the system throughput were not analyzed. Khan et al. [44] outlined that large scale deployments of networked IoT objects require CR capabilities, which leads to an emerging research dimension of CR-based IoT [45]. It becomes an enabling technology that can meet future IoT spectrum demands by dynamically utilizing the vacant channels while ensuring uninterrupted primary network communications. Embedding CR into IoT networks is complementary to existing efforts [46–48]. Dai et al. [49] proposed a cognitive multiple-input and single-output (MISO) precoding algorithm based on the signal-to-leakage-and-noise ratio (SLNR) criterion. The authors considered a scenario with a single secondary user, which can hardly be seen as an IoT network. Run et al. [50] proposed a new leakage-based precoding scheme for IoT communications with dimensional reduction and a SLNR criterion, thereby only analyzing the sum capacity and complexity. In [51] the use of MIMO for IoT connectivity to achieve high spectral efficiency, high data rates, and spatial multiplexing is discussed. However, MIMO for IoT is still a developing topic, as IoT connectivity has constraints and requirements that are significantly different to other broadband connections. D. Jiang et al. analyzed end-to-end network traffic reconstruction in large-scale networks where applying compressive sensing theory [52,53]. The authors analyzed the channel allocation problem in cognitive wireless networks for these medical applications [54]. Miah et al. [55] proposed cooperative spectrum sensing in a CRN using the EVD technique. However, the detection performance and the system throughput were not analyzed as with the MIMO antennas approach. Guilherme et al. [56] proposed the throughput of CRNs using eigenvalue-based cooperative spectrum sensing under complex Nakagami-m fading. However, the detection performance and the system throughput were not analyzed in a noise uncertainty environment with the MIMO antennas approach. Moreover, the authors could not utilize the weighted eigenvalue ratio. Ayse et al. [57] proposed the throughput analysis using eigenvalue-based spectrum sensing under noise uncertainty (i.e. SNR < -28). However, the authors could not utilize the WEVD ratio from the maximum eigenvalue to minimum eigenvalue.

In [58,59], the authors proposed the optimal channel sensing maximizing sum rate in CRNs with multiple secondary links. However, the system throughput was not analyzed in a noise uncertainty environment. The authors tried to maximize network throughput and minimize networks energy consumption by exploiting a network coding and sleeping scheme [60,61]. Miah et al. [62] proposed a maximization of the sum rate in amplify and forward (AF) cognitive radio networks using the superposition approach, where the sensing performance and the sum rate of CRNs were analyzed. However, the sensing and the system throughput were not analyzed in a noise uncertainty environment. Miah et al. [63] proposed an enhanced sum rate in a cluster-based CR relay network using the sequential approach to address future IoT requirements. However, both the sensing and the sum rate were not analyzed in a noise uncertainty environment and with the MIMO antennas approach. Moreover, both the energy consumption constraint and the expected lifetime were not analyzed either. The authors propose a robust routing algorithm to reach higher network energy efficiency, which is based on an optimization problem [64]. Vu et al. [65] proposed the throughput maximization for CRUs with energy constraints in an underlay paradigm, where the sensing performance, average throughput and energy consumption is analyzed using the conventional ED technique. However, the sensing performance, average throughput and energy consumption were not analyzed in a noise uncertainty environment with the MIMO antennas approach as the conventional ED technique was used, which is vulnerable to noise uncertainty. Chatterjee et al. [66] proposed a new protocol for concurrently allocating licensed spectrum to underlay cognitive users, where the error probability is being analyzed under the optimal threshold. Thakur et al. [67] proposed the performance analysis of CRNs using channel-prediction-probabilities and an improved frame structure, where the throughput and the error probability are analyzed in the overlay and underlay approaches.

3. The proposed MU-MIMO based CR-IoT system model

We propose a MU-MIMO based CR-IoT system model consisting of M CR-IoT users, each having T_r receive antennas and T_t transmit antennas; the FC has a single antenna. In this model, the CR-IoT user is denoted as an unlicensed user that opportunistically accesses the spectrum of the PU without causing interference, whilst the PU is denoted as the licensed user of the spectrum. The primary network (PN) consists of PU transmitters and PU receivers. We assume that the PU is always authorized to use the channel. The PU traffic across the channel is modeled by the discrete-time semi-Markov (two-state) process [68–70], as shown in Fig. 1. Without loss of generality, it is assumed that the duration of the $p_0 = OFF$ and $p_1 = ON$ states are represented by the random variables A and B, respectively. The random variables A and B follow exponential distribution with mean duration R and S. These

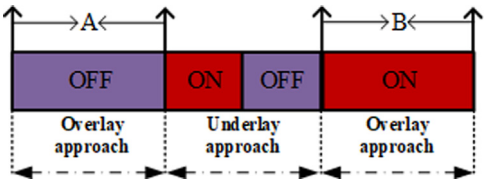


Fig. 1. Activity model of primary user (PU).

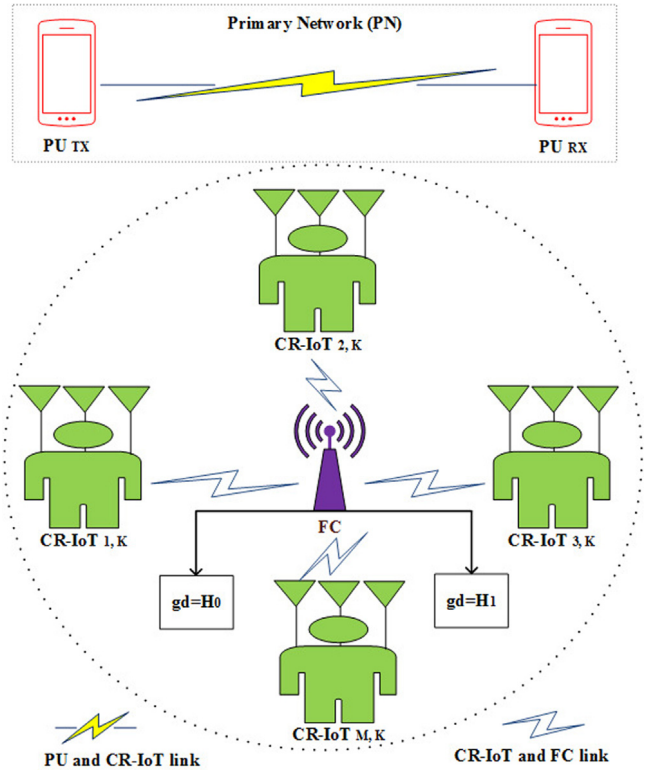


Fig. 2. The system model of the proposed MU-MIMO based CR-IoT scheme.

distributions are to be independent. At any time, the probabilities that the PU is in either the p_1 or p_0 state are given by $p_1 = S/S + R$, and $p_0 = R/R + S$, respectively. We consider that both the PN and CR-IoT network are perfectly synchronized, i.e. time-slotted, and the activity of the PU is changed only at the beginning of each time-slotted frame. As a PU periodically transmits packets including a pre-determined preamble, a CR-IoT can detect the starting point of the packet and synchronize with PN and sense the channel with the preamble of the PU packet. With this operation, the CR-IoT can synchronize with the PN [69].

Fig. 2 shows the general deployment of the CR-IoT network with two main sequential links: the primary user and cognitive radio IoT user (PU and CR-IoT User) link and secondly the cognitive radio IoT user and fusion center (CR-IoT and FC) link. For the simplicity of illustration, we assume that each CR-IoT user has an equal number of T_t transmit antennas and T_r receive antennas, i.e., $T_t = T_r$.

3.1. Characterization of primary user and cognitive radio IoT user (PU and CR-IoT user) link

In the PU and CR-IoT user link each CR-IoT user performs local spectrum sensing independently to detect PU activities. Let H_0 and H_1 be the hypotheses representing the absence and presence of a PU signal. For each CR-IoT user, the spectrum sensing problem can be formulated as follows:

$$\begin{cases} H_0: & \text{if PU is absent} \\ H_1: & \text{if PU is present} \end{cases} \tag{1}$$

Depending on the packet transmission of the PU, the received signals of the ith CR-IoT user with k antennas can be formulated as:

$$y_{i,k}(l) = \begin{cases} n_{i,k}(l); & H_0 \\ h_{i,k}(l)\, x(l) + n_{i,k}(l); & H_1 \end{cases} \tag{2}$$

where $i = 1, 2, 3, \ldots, M$, $k = 1, 2, 3, \ldots, K$ and $l = 1, 2, 3, \ldots, N_x$; here N_x is the total number of samples of the received signal that is defined as $N_x = 2\tau_s B$, where τ_s is the sensing time duration in seconds, and B is the prior known bandwidth in Hz. Here, $y_{i,k}(l)$ is the received signal of the ith CR-IoT user with k antennas, $x(l)$ is the transmitted signal of the PU with zero mean and variance σ_x^2, i.e. $x(l) \sim CN(0, \sigma_x^2)$, and $n_{i,k}(l)$ is a circularly symmetric complex Gaussian (CSCG) noise with variance $\sigma_{ni,k}^2$ of the ith CR-IoT user with k antennas, i.e. $n_{i,k}(l) \sim CN(0, \sigma_{ni,k}^2)$. Also, $h_{i,k}(l)$ is the channel gain of the ith CR-IoT user with k antennas. Finally, H_1 and H_0 denote the presence or absence of a PU respectively.

3.1.1. Conventional scheme with the Energy Detection (ED) technique

The conventional ED technique is widely used for the spectrum sensing of CR-IoT users, because it can be implemented easily without acquiring any prior information of the PU signal. The structure of the channel sensing process of the ith CR-IoT transmitter employing the ED is shown in Fig. 3.

The sensing result $y_{i,k}(l)$ received by the ith CR-IoT transmitter with k antennas is the signal power in a particular frequency in the time domain; a band-pass filter is applied to the received signal, then the output of this filter is transformed by an analog-to-digital converter (ADC), which will be individually averaged and squared to estimate its own measured energy $E_{i,k}$:

$$E_{i,k} = \frac{1}{N_x} \sum_{l=1}^{N_x} |y_{i,k}(l)|^2 \tag{3}$$

When N_x is relatively large, based on the central limit theorem (CLT), $E_{i,k}$ in (3) can be approximated as a Gaussian distribution under both hypotheses H_0 and H_1, with mean $(\mu_{0i,k}, \mu_{1i,k})$ and variance $\left(\sigma_{0i,k}^2, \sigma_{1i,k}^2\right)$ respectively, as follows [62,63]:

$$E_{i,k} = \begin{cases} \mu_{0i,k} = 2\tau_s B; & \sigma_{0i,k}^2 = 4\tau_s B \\ \mu_{1i,k} = 2\tau_s B\left(1 + \gamma_{i,k}\right); & \sigma_{1i,k}^2 = 4\tau_s B\left(1 + 2\gamma_{i,k}\right) \end{cases} \tag{4}$$

Here $\gamma_{i,k}$ is the signal-to-noise ratio (SNR) of the ith CR-IoT user with k antennas.

Considering that the local decision threshold at the ith CR-IoT user with k antennas is $\lambda_{i,k}$, the overall probability of false alarm, $p_{i,k}^f = Pr(H_1|H_0)$, and the probability of detection, $p_{i,k}^d = Pr(H_1|H_1)$, of the ith CR-IoT user with k antennas can be written as [62,63]:

$$p_{i,k}^f = Pr\left(E_{i,k} > \lambda_{i,k}|H_0\right) = Q\left(\frac{\lambda_{i,k} - \mu_{0i,k}}{\sqrt{\sigma_{0i,k}^2}}\right) = Q\left(\frac{\lambda_{i,k}}{\sigma_{0,i,k}^2} - 1\right)\sqrt{N_x(i,k)} \tag{5}$$

and

$$p_{i,k}^d = Pr\left(E_{i,k} > \lambda_{i,k}|H_1\right) = Q\left(\frac{\lambda_{i,k} - \mu_{1i,k}}{\sqrt{\sigma_{1i,k}^2}}\right) = Q\left(\frac{\lambda_{i,k}}{\sigma_{1,i,k}^2} - \gamma_{i,k} - 1\right)\sqrt{\frac{N_x(i,k)}{(1 + 2\gamma_{i,k})}} \tag{6}$$

where $Q(.)$ is the Gaussian tail function, which is defined as $Q(x) = \frac{1}{2\pi}\int_x^\infty exp(-\frac{t^2}{2})dt$.

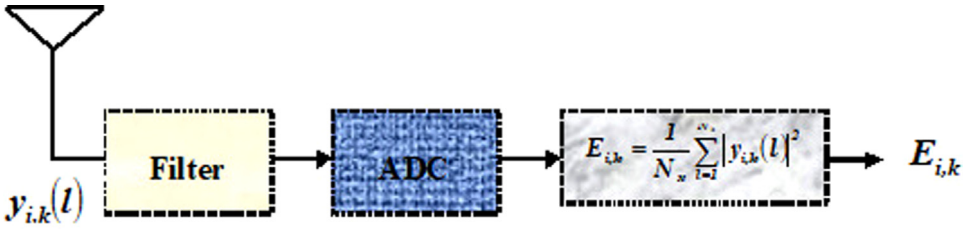


Fig. 3. Block diagram of the energy detection technique [63].

Proposition 1. *In the conventional scheme with the ED technique, the ith CR-IoT user with a single antenna ($K = 1$) can obtain a fixed number of samples, $N_x^c(i,1)$ which is defined as follows:*

$$N_x^c(i,1) = \frac{1}{\gamma_{i,1}^2}\left[Q^{-1}\left(p_{i,1}^f\right) - Q^{-1}\left(p_{i,1}^d\right)\sqrt{(1+2\gamma_{i,1})}\right] \quad \square \tag{7}$$

3.1.2. Conventional scheme with the Eigenvalue-Based Detection (EVD) technique

Let us consider a consecutive smoothing factor F. From Eq. (2), a set of signals $y_{i,k}(l)$ received by the ith CR-IoT user with k antennas can be written in the following matrix form under the H_0 hypothesis [55]:

$$\begin{bmatrix} y_{i,k}(l) \\ y_{i,k}(l-1) \\ \dots \\ \dots \\ y_{i,k}(l-F+1) \end{bmatrix} = \begin{bmatrix} n_{i,k}(l) \\ n_{i,k}(l-1) \\ \dots \\ \dots \\ n_{i,k}(l-F+1) \end{bmatrix} \tag{8}$$

Similarly, from Eq. (2), a set of the received signals $y_{i,k}(l)$ at the ith CR-IoT user with k antennas can be written in the following matrix form under the H_1 hypothesis [55]:

$$\begin{bmatrix} y_{i,k}(l) \\ y_{i,k}(l-1) \\ \dots \\ \dots \\ y_{i,k}(l-F+1) \end{bmatrix} = h_{i,k}(l)\begin{bmatrix} x(l) \\ x(l-1) \\ \dots \\ \dots \\ x(l-F+1) \end{bmatrix} + \begin{bmatrix} n_{i,k}(l) \\ n_{i,k}(l-1) \\ \dots \\ \dots \\ n_{i,k}(l-F+1) \end{bmatrix} \tag{9}$$

Now, we determine the statistical covariance matrices of the transmitted signal $x(l)$, the noise signal $n_{i,k}(l)$ and the received signal $y_{i,k}(l)$, respectively, as follows:

$$Cov_x = E[xx^T] \tag{10}$$

$$Cov_n = E[nn^T] \tag{11}$$

$$Cov_y = E[yy^T] \tag{12}$$

where

$$x = \begin{bmatrix} x(l) & x(l-1) & \dots & \dots & x(l-F+1) \end{bmatrix}^T \tag{13}$$

$$n = \begin{bmatrix} n_{i,k}(l) & n_{i,k}(l-1) & \dots & \dots & n_{i,k}(l-F+1) \end{bmatrix}^T \tag{14}$$

$$y = \begin{bmatrix} y_{i,k}(l) & y_{i,k}(l-1) & \dots & \dots & y_{i,k}(l-F+1) \end{bmatrix}^T \tag{15}$$

We can calculate the statistical covariance matrix (Cov_y) under the two hypotheses:

$$Cov_y = \begin{cases} \sigma_{0i,k}^2 I_F; & H_0 \\ Cov_x + \sigma_{0i,k}^2 I_F; & H_1 \end{cases} \tag{16}$$

where I_F is an identity matrix with size F, with F being a positive integer-valued smoothing factor.

Assuming ideal conditions, we calculate the sample autocorrelations of the received signal as follows:

$$A(q) = \frac{1}{N_x}\sum_{n=0}^{N_x-1} y_{i,k}(n)y_{i,k}(n-q) \tag{17}$$

Proposition 2. *In the conventional scheme with the EVD technique, the ith CR-IoT user with a single antenna ($K = 1$) obtains a fixed number of samples, $N_x^c(i,1)$ which is defined as follows:*

$$\begin{aligned} N_x^c(i,1) &= 2\tau_s B \\ &= iN_x \quad \square \end{aligned} \tag{18}$$

Proof. See Appendix B.

In a realistic scenario, the sample covariance matrix can only be calculated using a limited number of signal samples N_x^c as follows:

$$Cov_y(N_x^c) = \begin{bmatrix} A(0) & A(1) & \dots & A(F-1) \\ A(1) & A(0) & \dots & A(F-2) \\ \dots & \dots & \dots & \dots \\ A(F-1) & A(F-2) & \dots & A(0) \end{bmatrix} \tag{19}$$

Now we can calculate the decision statistics using Algorithm 1, where each CR-IoT user with a single antenna ($K = 1$) obtains the fixed number of samples N_x^c (see line 5). Then it computes the eigenvalue (see line 7) as well as the maximum eigenvalue ζ_{max} and the minimum eigenvalue ζ_{min} (see line 9). After that, it computes their ratio $r_{mtm}^c(i)$ (see line 9), and the local decision ld_i (see line 11). Finally, it computes the global detection gain, $gd_{f|d}$ (see line 12).

Algorithm 1 In the conventional scheme with the EVD technique, all CR-IoT users with a single antenna ($K = 1$) obtain a fixed number of samples, $N_x^c(i,1)$.

Input: Select appropriate smoothing factor, F, the number of samples, N_x and the global decision threshold, λ_{EVD}^c

Output: Calculate the global probability of detection, gd_d and the global probability of false alarm, gd_f

1: Initialize F, N_x, λ_{EVD}^c
2: **for** i from 1 to M **do**
3: **for** k from 1 to K **do**
4: **if** K==1 **then**
5: Set: $N_x^c(i,k) = iN_x$; Based on Proposition 2
6: **end if**
7: Calculate: the eigenvalue $\zeta = eign[Cov_y(N_x^c)]$
8: Update: $E(\zeta) = sorting(\zeta_1 < \zeta_2 < \zeta_3 < ...)$
9: Calculate: the maximum, ζ_{max} and the minimum, ζ_{min} from the eigenvalue, $E(\zeta)$ and their ratio, $r_{mtm}^c(i) = \frac{\zeta_{max}}{\zeta_{min}}$
10: **end for**
11: Set: $ld_i = r_{mtm}^c(i)$
12: Calculate: $gd_{f|d} = \begin{cases} 1, & \begin{cases} if \sum_{i=1}^{M} ld_i > \lambda_{EVD}^c; & OR-rule \\ if \sum_{i=1}^{M/2} ld_i > \lambda_{EVD}^c; & M-rule \end{cases} \\ 0, & Otherwise \end{cases}$
13: **end for**

3.1.3. Proposed scheme with the Weighted Eigenvalue-Based Detection (WEVD) technique

A major challenge for future IoT networks is an efficient spectrum utilization. The proposed scheme with the WEVD technique is a promising solution for the spectrum scarcity problem in noise uncertainty

environments; here each CR-IoT user is being equipped with MIMO antennas that obtain more samples N_x^p compared to both conventional schemes with the EVD and the ED technique. For example, the CR-IoT$_{1,2}$ user, the 1st CR-IoT user with 2 antennas, can obtain more samples, $N_x(1,2)$ compared to the 1st CR-IoT user with 1 antenna for sensing the PUs signal, i.e., $N_x(1,2) > N_x(1,1)$. Similarly, for the CR-IoT$_{1,3}$ user, the 1st CR-IoT user with 3 antennas can obtain more number of samples, $N_x(1,3)$, compared with the 1st CR-IoT user with 1 antenna and 2 antennas, respectively, i.e. $N_x(1,3) > N_x(1,2) > N_x(1,1)$ and so on. Fig. 4 shows the frame structure of the proposed MU-MIMO based CR-IoT scheme. Here each frame distinguishes between two phases, (i) the setup phase and (ii) the steady state phase.

In the setup phase, each CR-IoT user sends information about its ID, current location and SNR of reporting channel to the FC. After that the FC broadcasts the sensing request to each CR-IoT user, and then all CR-IoT users sense with k antennas the PU channel during the fixed sensing time slot τ_s. Afterwards each CR-IoT user makes a local decision, which is forwarded to the corresponding FC during a fixed reporting time slot, τ_r, through a common control channel. After that the FC makes a global decision which is broadcast to the entire CR-IoT network.

In the steady state phase, after getting a global decision from the FC, the CR-IoT transmitter users send their data to the corresponding CR-IoT receiver users during the fixed transmission time slot $T-\tau_s-\tau_r$.

Proposition 3. *In the proposed MU-MIMO based CR-IoT scheme with the WEVD technique, the ith CR-IoT user with k antennas obtains a non-fixed number of samples $N_x^p(i,k)$ which is defined as follows:*

$$\begin{aligned} N_x^p(i,k) &= 2iK\tau_s B \\ &= ikN_x \quad \square \end{aligned} \tag{20}$$

Proof. See Appendix C.

We can rewrite the covariance matrix based on a non-fixed number of samples, N_x^p using Eq. (19) as follows:

$$\overline{Cov_y}(N_x^p) = \begin{bmatrix} \overline{A}(0) & \overline{A}(1) & \dots & \overline{A}(F-1) \\ \overline{A}(1) & \overline{A}(0) & \dots & \overline{A}(F-2) \\ \dots & \dots & \dots & \dots \\ \overline{A}(F-1) & \overline{A}(F-2) & \dots & \overline{A}(0) \end{bmatrix} \tag{21}$$

Now we can calculate the decision statistics using Algorithm 2. Here each CR-IoT user with multiple antennas ($k = K$) obtains a non-fixed number of samples N_x^p (see line 4). Then the algorithm computes the eigenvalue (see line 5) and also the maximum eigenvalue $\overline{\zeta}_{max}$ as well as the minimum eigenvalue $\overline{\zeta}_{min}$ (see line 7). After that, it computes their ratio $\overline{r}_{mtm}^p(i)$ and the weight ratio ω_i (see line 8), which is defined as the ratio between the difference of $\overline{\zeta}_{max}$ and $\overline{\zeta}_{min}$ to the sum of $\overline{\zeta}_{max}$ and $\overline{\zeta}_{min}$; it then computes the ith local decision $\overline{ld}_i$ (see line 10). Finally, it computes the global detection gain $\overline{gd}_{f|d}$ (see line 11), the average throughput $R_{avg}\left(N_x^p\right)$ (see line 12), the average energy consumption $E_{avg}\left(N_x^p\right)$ (see line 13), and the expected life time, η (see line 14).

3.2. Characterization of the CR-IoT user and FC link

We assume that all CR-IoT users communicate with a fixed FC through a dedicated control channel. Each CR-IoT user makes its own local decision on the presence of the PU signal based on its own spectrum sensing, and this local decision is forwarded to the FC, which makes a final global decision using a HDF rule such as OR logic ("OR-rule"), Majority logic ("M-rule"), etc.

Algorithm 2 In the proposed a MU-MIMO based CR-IoT scheme with the WEVD technique, all CR-IoT users with the kth antenna have obtained the non-fixed number of samples N_x^p.

Input: Select appropriate smoothing factor F, the number of samples N_x, and the global decision threshold λ_{WEVD}^p

Output: Calculate the global probability of detection $\overline{gd}_d$, and the global probability of false alarm $\overline{gd}_f$

1: Initialize F, N_x, λ_{WEVD}^p
2: **for** i from 1 to M **do**
3: **for** k from 1 to K **do**
4: Set: $N_x^p(i,k) = ikN_x$; Based on Proposition 3
5: Calculate: the Eigenvalue $\overline{\zeta} = eign[\overline{Cov_y}(N_x^p)]$
6: Update: $E(\overline{\zeta}) = sorting(\overline{\zeta_1} < \overline{\zeta_2} < \overline{\zeta_3} < ...)$
7: Calculate: the maximum, $\overline{\zeta}_{max}$ and the minimum, $\overline{\zeta}_{min}$ from the Eigenvalue, $E(\overline{\zeta})$ and their ratio, $\overline{r}_{mtm}^p(i) = \frac{\overline{\zeta}_{max}}{\overline{\zeta}_{min}}$
8: Calculate: the weight, $\omega_i = \frac{\overline{\zeta}_{max}-\overline{\zeta}_{min}}{\overline{\zeta}_{max}+\overline{\zeta}_{min}}$
9: **end for**
10: Set: $\overline{ld}_i = \omega_i \overline{r}_{mtm}^p(i)$
11: Calculate the detection gain:
$$\overline{gd}_{f|d} = \begin{cases} 1, & \begin{cases} if \sum_{i=1}^{M} \overline{ld}_i > \lambda_{WEVD}^p; & OR-rule \\ if \sum_{i=1}^{M/2} \overline{ld}_i > \lambda_{WEVD}^p; & M-rule \end{cases} \\ 0, & Otherwise \end{cases}$$
12: Calculate the average throughput:
$R_{avg}\left(N_x^p\right) = \left(\frac{T-\tau_s}{T}\right)\left(c_0(1-\overline{gd}_f)p_0 + c_1(1-\overline{gd}_d)p_1\right)$; Based on Eq. (25)
13: Calculate the average energy consumption:
$E_{avg}\left(N_x^p\right) = e_s\tau_s + e_t t_t\left((1-\overline{gd}_f)p_0 + (1-\overline{gd}_d)p_1\right)$; Based on Eq. (28)
14: Calculate the expected life time:
$\eta = \frac{e_c}{E_{avg}(N_x^p)}$; Based on Eq. (29)
15: **end for**

3.2.1. Conventional scheme with the energy detection (ED) technique

In the conventional scheme with the ED technique the final global probability of detection gd_{ED}^d and the global probability of false alarm gd_{ED}^f of the entire CR-IoT network at the FC are given as follows [55]:

$$\begin{cases} gd_{ED}^d = 1 - \prod_{i=1}^{M}(1-p_{i,k}^d)(1-p_{i,k}^e) + p_{i,k}^d p_{i,k}^e; & OR-rule \\ gd_{ED}^f = 1 - \prod_{i=1}^{M}(1-p_{i,k}^f)(1-p_{i,k}^e) + p_{i,k}^f p_{i,k}^e; & OR-rule \\ gd_{ED}^d = 1 - \prod_{i=1}^{M/2}(1-p_{i,k}^d)(1-p_{i,k}^e) + p_{i,k}^d p_{i,k}^e; & M-rule \\ gd_{ED}^f = 1 - \prod_{i=1}^{M/2}(1-p_{i,k}^f)(1-p_{i,k}^e) + p_{i,k}^f p_{i,k}^e; & M-rule \end{cases} \tag{22}$$

where $p_{i,k}^e$ is the reporting error probability at the ith CR-IoT user with k antennas.

3.2.2. Conventional scheme with the EVD technique

In the conventional scheme with the EVD technique, the final global decision $gd_{f|d}$ of the entire CR-IoT network at the FC is made upon using the two HDF rules as follows [55]:

$$gd_{f|d} = \begin{cases} 1, & \begin{cases} if \sum_{i=1}^{M} ld_i > \lambda_{EVD}^c; & OR-rule \\ if \sum_{i=1}^{M/2} ld_i > \lambda_{EVD}^c; & M-rule \end{cases} \\ 0, & Otherwise \end{cases} \tag{23}$$

where λ_{EVD}^c is the global decision threshold of the conventional scheme with the EVD technique.

3.2.3. Proposed scheme with the Weighted Eigenvalue-Based Detection (WEVD) technique

In the proposed scheme with the WEVD technique, the final global decision $\overline{gd}_{f|d}$ of the entire CR-IoT network is made at the FC upon

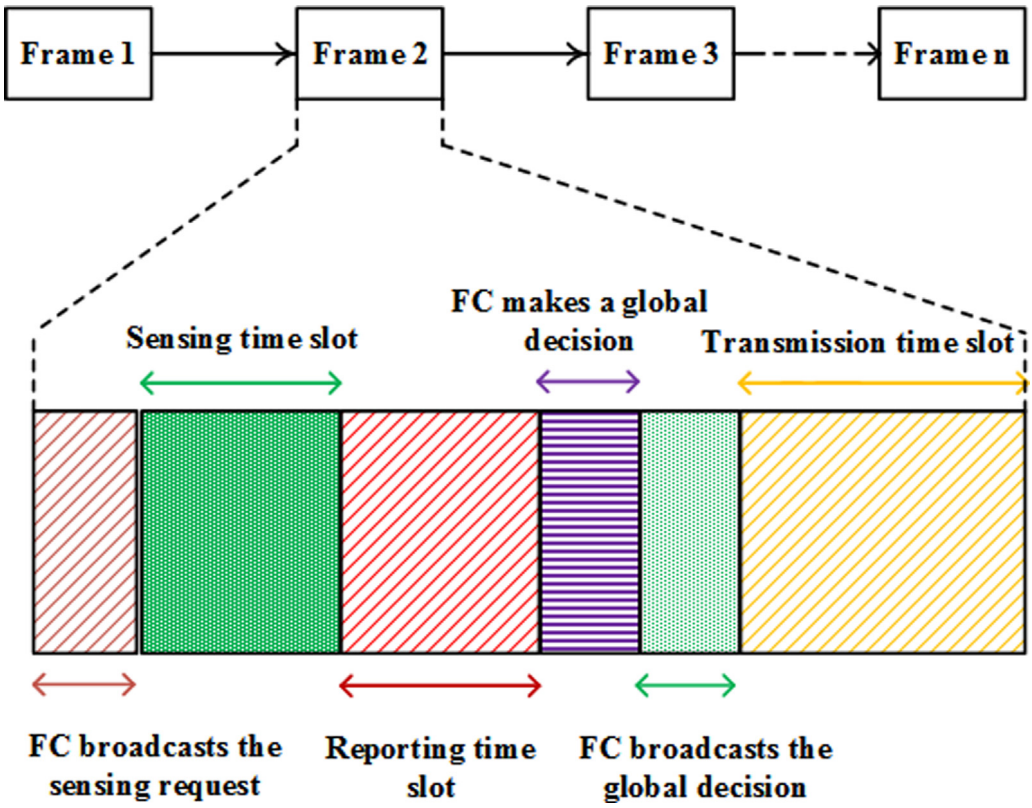


Fig. 4. Frame structure of a time slot for sensing, reporting and packet transmission in the proposed MU-MIMO based CR-IoT scheme with the WEVD technique.

using the two HDF rules as follows:

$$\overline{gd}_{f|d} = \begin{cases} 1, & \begin{cases} if \sum_{i=1}^{M} \overline{ld}_i > \lambda_{WEVD}^{p}; & OR-rule \\ if \sum_{i=1}^{M/2} \overline{ld}_i > \lambda_{WEVD}^{p}; & M-rule \end{cases} \\ 0, & Otherwise \end{cases} \tag{24}$$

where λ_{WEVD}^{p} is the global decision threshold of the proposed scheme with the WEVD technique.

4. Throughput and energy constraint analysis of the proposed scheme

For the proposed scheme with the WEVD technique the system throughput is discussed in the section "Throughput Analysis", while the average energy consumption and the expected lifetime are discussed in the section "Energy Constraint Analysis". Moreover, the time complexity is discussed in the section "Complexity Analysis".

4.1. Throughput analysis

If the CR-IoT user transmitter does not detect the PU signal in the transmission slot, it determines that the channel is available and transmits data to its receiver; otherwise, it waits until the channel becomes available for its transmission, which is scheduled based on a round-robin protocol [58]. In the round-robin protocol, all CR-IoT users check the availability of the channel and report their sensing results to a FC managing the CR-IoT users with the out-of-band control channel. When a CR-IoT user detects the busy channel occupied by a PU, the CR-IoT user does not send their packets and wait until the next time slot. If all CR-IoT users detect the idle channel, they can transmit packets one by one under the control of the FC. When the PU is absent and the absence of the PU is perfectly detected by the CR-IoT user, the CR-IoT user can access the primary spectrum with the probability $(1 - \overline{gd}_f)$; otherwise, when the PU is present and the presence of the PU is perfectly detected by the CR-IoT user, the CR-IoT user cannot access the primary spectrum with the probability $\overline{gd}_d$. In this case, the average throughput of all PU and CR-IoT users, that access the network in a round-robin manner, is calculated as follow [62,63,65]:

$$R_{avg}\left(N_x^p\right) = \left(\frac{T - \tau_s - \tau_r}{T}\right)\left(c_0\left(1 - \overline{gd}_f\right)p_0 + c_1\left(1 - \overline{gd}_d\right)p_1\right) \tag{25}$$

where c_0 denotes the channel capacity of the CR-IoT user transmitter and the CR-IoT user receiver link when the PU is absent, c_1 denotes the channel capacity of the CR-IoT user transmitter and CR-IoT user receiver link when the PU is present, T is the length of a time slot, p_0 and p_1 are the probability of absence and presence of the PU signal, respectively.

When the CR-IoT user transmits data when the PU is absent, the CR-IoT user transmitter and the CR-IoT user receiver link can achieve the following throughput:

$$c_0 = log_2\left(1 + SNR_{CR-IoT}\right) \tag{26}$$

where SNR_{CR-IoT} is the SNR in the CR-IoT user transmitter and the CR-IoT user receiver link.

When the CR-IoT user receiver receives data when the PU is present, the CR-IoT user receiver receives the PU signal as noise. In this case, the CR-IoT user transmitter and the CR-IoT user receiver link can achieve the following throughput:

$$c_1 = log_2\left(1 + \frac{SNR_{CR-IoT}}{1 + SNR_{PU}}\right) \tag{27}$$

where SNR_{PU} is the SNR in the PU transmitter and the CR-IoT user receiver link.

Proposition 4. *In the proposed scheme with the WEVD technique, the appearance of the PU signal can reduce the achievable throughput of the CR-IoT user transmitter and the CR-IoT user receiver link, i.e.* $c_0 > c_1$. □

Table 2
Simulation parameters.

Parameters	Value
The number of samples, N_x	60
The sensing time, τ_s	1 ms
The time slot length, T	10 ms
The channel bandwidth, B	300 kHz
The number of CR-IoT users, M	4
The number of PU transmitters, PU_{TX}	1
The number of PU receivers, PU_{RX}	1
The smoothing factor, F	4
The PU signal, $x(t)$	BPSK
The channel noise in CR-IoT user, $n(t)$	AWGN
The SNR of each PU user, SNR_{PU}	10 dB
The SNR of each CR-IoT user, SNR_{CR-IoT}	−28 dB
The energy consumed for the sensing duration, e_s	1 J
The energy consumed for the transmission duration, e_t	3 J
The capacity of the battery, e_c	300 J
The number of rounds	150
The probability of the absence of the PU, p_0	0.5
The probability of the presence of the PU, p_1	0.5

4.2. Energy constraint analysis

In this section, we can calculate the average energy consumption of the proposed scheme with the WEVD technique as follows [65]:

$$E_{avg}\left(N_x^p\right) = e_s\tau_s + e_t t_t\left((1-\overline{gd}_f)p_0 + (1-\overline{gd}_d)p_1\right) \tag{28}$$

where t_t is the transmission time which is defined as $t_t = T - \tau_s - \tau_r$, e_s is the energy consumed for the sensing duration, T is the time slot duration and e_t is the energy consumed for the transmission.

The expected lifetime of the proposed scheme with the WEVD technique can be calculated as follows [65]:

$$\eta = \frac{e_c}{E_{avg}\left(N_x^p\right)} \tag{29}$$

where e_c is the capacity of battery.

4.3. Complexity analysis

In this subsection, we analyze the complexity of the conventional scheme based Algorithm 1 and the proposed scheme based Algorithm 2.

Algorithm 1: The total time complexity of the conventional scheme ($K = 1$) based on Algorithm 1 is calculated as follows:

$$\sum_{i=1}^{M}\sum_{k=1}^{K} C_0 = C_0 \times K \times M = C_0 \times M \tag{30}$$

where C_0 is a constant that denotes the computing cost of instructions from lines 2–13. Therefore, the Big O time complexity of the conventional Algorithm 1 is $O(M)$.

Algorithm 2: The total time complexity of the proposed scheme ($K = 2$), i.e., K is a constant based on Algorithm 2 is calculated as follows:

$$\sum_{i=1}^{M}\sum_{k=1}^{K} C_1 = C_1 \times K \times M = C_1 \times M \tag{31}$$

where C_1 is a constant that denotes the computing cost of instructions from lines 2–15. Therefore, the Big O time complexity of the proposed scheme based on Algorithm 2 is $O(M)$.

5. Simulation results and discussion

In this section we verify the theoretical results and evaluate the detection gain and the system throughput of the proposed scheme with the WEVD technique through Monte Carlo simulation using MATLAB, which was carried out under the following conditions as shown in Table 2.

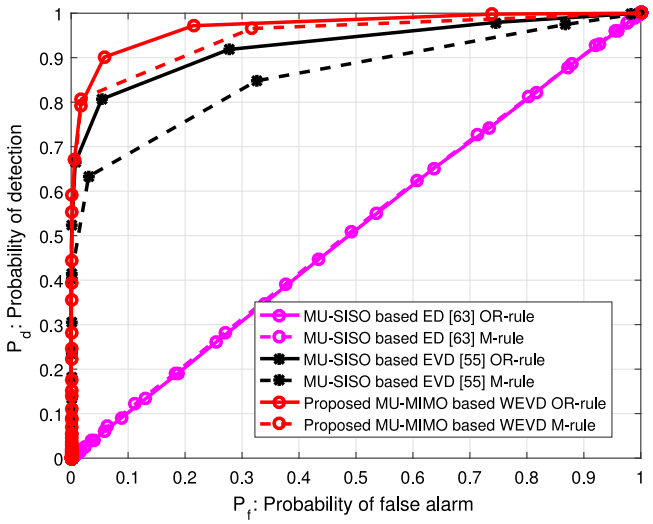


Fig. 5. The ROC curves of the conventional MU-SISO based CR-IoT scheme with the ED technique [63], the conventional MU-SISO based CR-IoT scheme with the EVD technique [55] and the proposed MU-MIMO based CR-IoT scheme with the WEVD technique, here $T_t = 2$, $T_r = 2$ and $N_x = 60$.

Table 3
Probability of detection vs. the number of samples with a probability of false alarm of 0.1.

Schemes	$N_x = 40$	$N_x = 50$	$N_x = 60$
Conventional scheme with the ED technique	$gd^d_{ED} = 0.1$	$gd^d_{ED} = 0.1$	$gd^d_{ED} = 0.1$
Conventional scheme with the EVD technique	$gd_d = 0.60$	$gd_d = 0.64$	$gd_d = 0.79$
Proposed scheme with the WEVD technique	$\overline{gd}_d = 0.84$	$\overline{gd}_d = 0.88$	$\overline{gd}_d = 0.92$

Under the conditions shown in Table 2 the curves of receiver operating characteristics (ROC) are illustrated in Fig. 5. It is shown that the proposed MU-MIMO based CR-IoT scheme with the WEVD technique achieves better detection gain compared to both the conventional MU-SISO based CR-IoT scheme with the ED technique and the conventional MU-SISO based CR-IoT scheme with the EVD technique. In the proposed scheme, each CR-IoT user is being equipped with MIMO antennas such as $T_t = 2$, $T_r = 2$, which obtain more samples as shown in Table 3. The proposed scheme utilizes the WEVD ratio, which is defined as the ratio between the difference of the maximum eigenvalue, $\overline{\zeta}_{max}$, and the minimum eigenvalue, $\overline{\zeta}_{min}$, to the sum of $\overline{\zeta}_{max}$ and $\overline{\zeta}_{min}$. Moreover, the probability of detection of the "OR-rule" is always larger compared to the "M-rule", except for the conventional MU-SISO based CR-IoT scheme with the ED technique; here it is easy to see that the performances of the "OR-rule" and the "M-rule" are similar in a noise uncertainty environment. Therefore, the proposed scheme with the WEVD technique outperforms the other schemes for all cases, which makes it more applicable for future IoT networks.

When comparing the sensing gain at the FC, it can be seen in Table 3 that the proposed MU-MIMO based CR-IoT scheme with the WEVD technique can detect the spectrum with 92% detection accuracy, whereas the conventional scheme with the EVD technique and the conventional scheme with the ED technique only detect the PU's signal with 79% and 10% detection accuracy, respectively. Therefore, the proposed scheme with the WEVD technique is a better solution to address the scarcity problem for the future IoT networks.

Fig. 6 shows the system throughput for the conventional scheme with the ED technique, the conventional scheme with the EVD technique and the proposed scheme with the WEVD technique. The system throughput is a function of the probability of false alarm, $\overline{gd}_f$. The

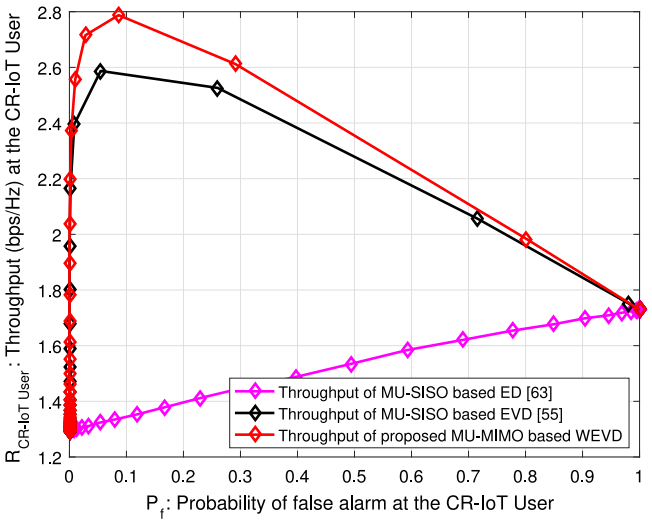


Fig. 6. The system throughput curves vs. the probability of false alarm at the CR-IoT user in the conventional MU-SISO based CR-IoT scheme with the ED technique [63], the conventional MU-SISO based CR-IoT scheme with the EVD technique [55] and the proposed MU-MIMO based CR-IoT scheme with the WEVD technique.

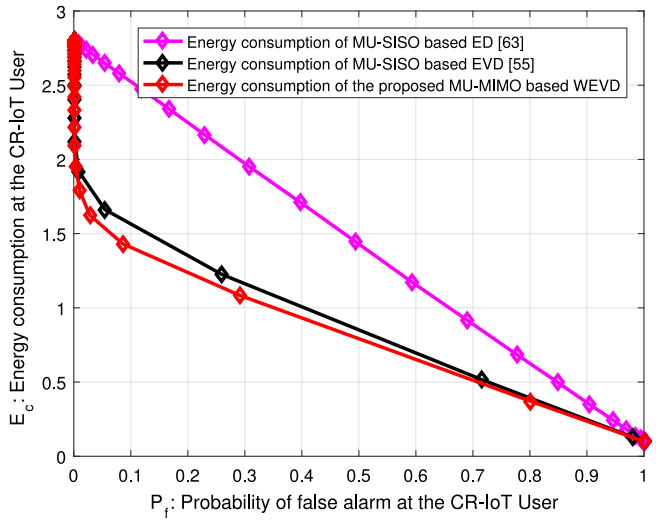


Fig. 7. The energy consumption curves vs. the probability of false alarm at the CR-IoT user in the conventional MU-SISO based CR-IoT network with the ED technique [63], the conventional MU-SISO based CR-IoT network with the EVD technique [55] and the proposed MU-MIMO based CR-IoT network with the WEVD technique.

Table 4
Throughput of the different schemes when the number of samples under the probability of false alarm is 0.1.

Schemes	$N_x = 40$	$N_x = 50$	$N_x = 60$
Throughput of the conventional scheme with the ED technique	1.39	1.39	1.39
Throughput of the conventional scheme with the EVD technique	2.18	2.41	2.60
Throughput of the proposed scheme with the WEVD technique	2.62	2.78	2.80

system throughput of the proposed scheme with the WEVD technique is higher compared to both the conventional scheme with the ED technique and the conventional scheme with the EVD technique.

Table 4 shows that the proposed scheme with the WEVD technique achieves a system throughput of 2.80 Hz compared to the conventional scheme with the EVD technique and the conventional scheme with the ED technique 2.60 Hz and 1.39 Hz, respectively. This enhanced system throughput makes it more applicable for future IoT networks than the two conventional schemes.

Fig. 7 shows the energy consumption for the conventional scheme with the ED technique, the conventional scheme with the EVD technique, and the proposed scheme with the WEVD technique. The average energy consumption E_{avg} is a function of the probability of false alarm, $\overline{gd}_f$. The average energy consumption of the proposed scheme with the WEVD technique has the best performance in comparison with the other conventional schemes due to its higher detection probability. As mentioned previously, the proposed scheme with the WEVD technique is better than the conventional scheme with the EVD technique when the probability of false alarm, $\overline{gd}_f > 0.7$.

The proposed scheme with the WEVD technique obtains an energy consumption of 1.35 compared to 1.50 and 2.50 with the conventional schemes, as shown in Table 5, making it more suitable for energy-constrained IoT devices.

Fig. 8 compares the expected lifetime of the conventional scheme with the ED technique, the conventional scheme with the EVD technique and the proposed scheme with the WEVD technique. Here, the proposed scheme with the WEVD technique enhances energy efficiency compared to both conventional schemes.

With regard to the expected lifetime, the proposed scheme with the WEVD technique achieves a system throughput of 2.80 Hz compared to 2.60 Hz (conventional scheme with the EVD technique) and 1.6 Hz

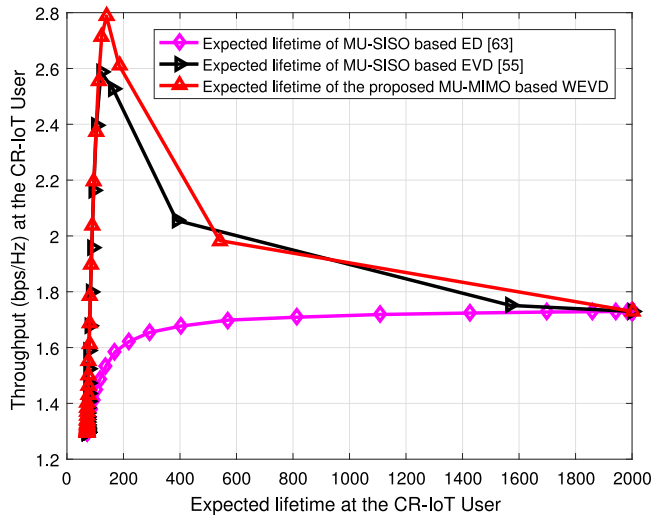


Fig. 8. The system throughput curves vs. the expected lifetime of a CR-IoT user with the conventional MU-SISO based CR-IoT network with the ED technique [63], the conventional MU-SISO based CR-IoT network with the EVD technique [55] and the proposed MU-MIMO based CR-IoT network with the WEVD technique when $e_c = 200$, $e_s = 1$ and $e_t = 3$.

Table 5
Comparing energy consumption of ED, EVD and WEVD when the number of samples under the probability of false alarm is 0.1.

Schemes	$N_x = 40$	$N_x = 50$	$N_x = 60$
Energy consumption of the conventional scheme with the ED technique	2.50	2.50	2.50
Energy consumption of the conventional scheme with the EVD technique	1.82	1.55	1.50
Energy consumption of the proposed scheme with the WEVD technique	1.50	1.48	1.35

(conventional scheme with the ED technique), as shown in Table 6. Therefore, the proposed scheme with the WEVD technique can prolong the expected lifetime compared to both conventional schemes, which is applicable for future IoT networks.

Table 6
Expected lifetime of ED, EVD and WEVD when the number of samples under the probability of false alarm is 0.1.

Schemes	$N_x = 40$	$N_x = 50$	$N_x = 60$
Expected lifetime of the conventional scheme with the ED technique	1.60	1.60	1.60
Expected lifetime of the conventional scheme with the EVD technique	2.20	2.48	2.60
Expected lifetime of the proposed scheme with the WEVD technique	2.70	2.78	2.80

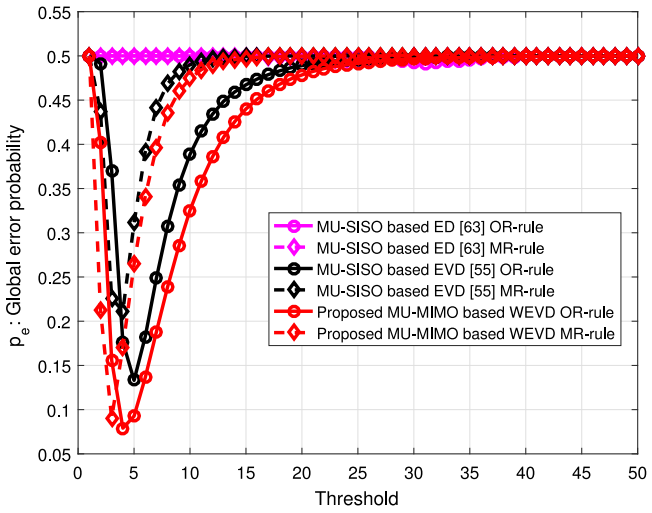


Fig. 9. The global error probability vs. the threshold at the FC in the conventional MU-SISO based CR-IoT network with the ED technique [63], the conventional MU-SISO based CR-IoT network with the EVD technique [55] and the proposed MU-MIMO based CR-IoT network with the WEVD technique when $p_1 = p_0 = 0.5$, $M = 4$ and $K = 4$.

We can calculate the global error probability p_e at the FC, which is given as follows [66,67]:

$$p_e = p_0 \overline{gd}_f + p_1 \left(1 - \overline{gd}_d\right) \tag{32}$$

Fig. 9 shows that p_e for both the conventional scheme with the EVD technique and the proposed scheme with the WEVD technique decreases along with an increase of the threshold value from 0 to 5. In addition, p_e for both the conventional scheme with the EVD technique and the proposed scheme with the WEVD technique gets larger along with an increase of the threshold value from 5 to 35. On the other hand, the global error probability p_e in the conventional scheme with the ED technique is high along with an increase of the threshold value from 0 to 50, here we consider a noise uncertainty environment with a low SNR regime, i.e. -28 dB. We can see that the global error probability p_e of the proposed scheme with the WEVD technique is lower than for both the conventional scheme with the EVD technique and the conventional scheme with the ED technique.

With regard to the global error probability, the proposed scheme with the WEVD technique achieves a global error probability of 0.025 compared to 0.14 with the conventional scheme with the EVD technique and 0.50 with the conventional scheme with the ED technique (Table 7). The proposed scheme achieves the lowest global error probability compared to both conventional schemes with threshold values from 0 to 50.

The results listed in Tables 3, 4, 5, 6 and 7 show that the proposed scheme with the WEVD technique achieves better sensing gain, enhanced system throughput, lower energy consumption, prolongs the expected lifetime and lower error probability compared to the conventional scheme with the EVD technique and the conventional scheme with the ED technique, making it more applicable for future IoT networks.

6. Conclusion and future work

In this paper we calculate the WEVD ratio, which is defined as the ratio between the difference of the maximum eigenvalue and minimum eigenvalue to the sum of the maximum eigenvalue and minimum eigenvalue in the proposed scheme, with the WEVD technique achieving outstanding detection gain and throughput performance. In the detection gain, the proposed scheme with the WEVD technique demonstrates a 13% and 82% improvement over the conventional scheme with the EVD technique and the conventional scheme with the ED technique, respectively. Also, with respect to system throughput, the proposed scheme with the WEVD technique is 7.14% and 50.35% better than the conventional scheme with the EVD technique and the conventional scheme with the ED technique, respectively. Moreover, the proposed scheme with the WEVD technique demonstrates for the number of samples ($N_x^p = 60$) a 15% and 46% lower energy consumption compared to both the conventional scheme with the EVD technique and the conventional scheme with the ED technique, respectively. With regard to the expected lifetime in terms of system throughput, the proposed scheme for round 150 shows a 7.14% and 42% improvement compared to both the conventional scheme with the EVD technique and the conventional scheme with the ED technique. Finally, a global error probability of 2.5% is achieved in the proposed scheme, while 14% and 50% global error probabilities are achieved in the conventional scheme with the EVD technique and the conventional scheme with the ED technique. Therefore, we conclude that our proposed scheme with the WEVD technique will be more applicable for future IoT networks due to the fact that it mitigates against the spectrum scarcity problem, enhances system throughput, improves energy efficiency and prolongs the expected lifetime.

Our future work will analyze the sensing characteristics and the throughput of MU-MIMO based CR-IoT networks that use the underlay approach.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

M.S. Miah: Conceptualization, Methodology, Software, Validation, Writing - original draft. **M. Schukat:** Supervision, Validation, Writing - reviewing & editing. **E. Barrett:** Supervision, Validation, Writing - reviewing & editing.

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Appendix A

Proof of Proposition 1. We focus on the BPSK signal and CSCG noise. The probability of a false alarm can be obtained by using Eq. (5):

$$\left(\frac{\lambda_{i,k}}{\sigma^2_{0i,k}} - 1\right)\sqrt{N_x(i,k)} = Q^{-1}\left(p^f_{i,k}\right) \tag{A.1}$$

Using Eq. (6) the probability of detection can be obtained as follows:

$$\left(\frac{\lambda_{i,k}}{\sigma^2_{i,k}} - \gamma_{i,k} - 1\right)\sqrt{\frac{N_x(i,k)}{(1+2\gamma_{i,k})}} = Q^{-1}\left(p^d_{i,k}\right) \tag{A.2}$$

Table 7
Comparing global error probability of ED, EVD and WEVD with threshold values from 0 to 50.

Schemes	$\lambda = 5$	$\lambda = 10$	$\lambda = 15$	$\lambda = 20$	$\lambda = 30$
The global error probability of the conventional scheme with the ED technique	0.50	0.50	0.50	0.50	0.50
The global error probability of the conventional scheme with the EVD technique	0.14	0.25	0.4	0.45	0.49
The global error probability of the proposed scheme with the WEVD technique	0.025	0.19	0.35	0.41	0.47

By substituting Eq. (A.2) into Eq. (A.1) and rewriting this equations, we have

$$\left(\frac{Q^{-1}\left(p^{f}_{i,k}\right)}{\sqrt{N_x(i,k)}} - \gamma_{i,k}\right)\sqrt{N_x} = Q^{-1}\left(p^{d}_{i,k}\right)\sqrt{(1+2\gamma_{i,k})}$$

$$Q^{-1}\left(p^{f}_{i,k}\right) - \gamma_{i,k}\sqrt{N_x(i,k)} = Q^{-1}\left(p^{d}_{i,k}\right)\sqrt{(1+2\gamma_{i,k})}$$

$$\gamma_{i,k}\sqrt{N_x(i,k)} = Q^{-1}\left(p^{f}_{i,k}\right) - Q^{-1}\left(p^{d}_{i,k}\right)\sqrt{(1+2\gamma_{i,k})}$$

$$\sqrt{N_x(i,k)} = \frac{1}{\gamma_{i,k}}\left[Q^{-1}\left(p^{f}_{i,k}\right) - Q^{-1}\left(p^{d}_{i,k}\right)\sqrt{(1+2\gamma_{i,k})}\right]$$

$$N^{c}_{x}(i,1) = \frac{1}{\gamma^{2}_{i,1}}\left[Q^{-1}\left(p^{f}_{i,1}\right) - Q^{-1}\left(p^{d}_{i,1}\right)\sqrt{(1+2\gamma_{i,1})}\right]^{2} \quad (A.3)$$

Therefore, the ith CR-IoT user with a single antenna in the conventional scheme with the ED technique has obtained fewer samples N^{c}_{x} in a noise uncertainty environment. Therefore, the detection performance is very poor, making it less applicable for the future IoT networks. □

Appendix B

Proof of Proposition 2. Each CR-IoT user in the conventional scheme with the EVD technique is being equipped with a single antenna ($K = 1$) which obtains a fixed number of samples. Here the sensing time slot τ_s and the bandwidth of the PU signal B are fixed, therefore the number of samples N^{c}_{x} is also a constant as follows:

$$\begin{aligned} N^{c}_{x} &= 2i\tau_s B \\ &= iN_x \end{aligned} \quad (B.1)$$

Based on this fixed number of samples, N^{c}_{x}, the detection performance in the conventional scheme with the EVD technique is poor. □

Appendix C

Proof of Proposition 3. From Fig. 2 it is observed that the ith CR-IoT user with 1 antenna has a non-fixed number of samples, $N^{p}_{x}(i,1)$ to sense the PUs signal as follows:

$$\begin{aligned} N^{p}_{x}(i,1) &= 2i\tau_s B \\ &= iN_x \end{aligned} \quad (C.1)$$

Likewise, for the ith CR-IoT user with 2 antennas, the number of samples $N^{p}_{x}(i,2)$ can be obtained as follows:

$$\begin{aligned} N^{p}_{x}(i,2) &= 4i\tau_s B \\ &= i2N_x \end{aligned} \quad (C.2)$$

In addition, for the ith CR-IoT user with k antennas, the number of samples $N^{p}_{x}(i,k)$ can be obtained as follows:

$$\begin{aligned} N^{p}_{x}(i,k) &= 2ik\tau_s B \\ &= ikN_x \end{aligned} \quad (C.3)$$

In summary, from the above equations, it is clear that the proposed scheme with the WEVD technique obtains a non-fixed number of samples N^{p}_{x} due to the increased index of antennas K, and we have

$$N^{p}_{x}(i,1) < N^{p}_{x}(i,2) < N^{p}_{x}(i,3) < \cdots < N^{p}_{x}(i,K) \quad (C.4)$$

Therefore, the ith CR-IoT user with k antennas in the proposed scheme with the WEVD technique has obtained more samples N^{p}_{x}, i.e. $N^{p}_{x} = kN^{c}_{x}$, based on both Eqs. (B.1) and (C.3). Due to this the detection performance in the proposed scheme with the WEVD technique is much better compared to both conventional schemes based on the EVD and the ED techniques, making it therefore more applicable for future IoT networks. □

Appendix D

Proof of Proposition 4. The first derivations of the average throughput, $R_{avg}\left(N^{p}_{x}\right)$ in Eq. (25) with respect to the global probability of false alarm, $\overline{gd}_f$ is given as follows:

$$\begin{aligned} \frac{\partial}{\partial \overline{gd}_f} R_{avg}\left(N^{p}_{x}\right) &= \frac{T-\tau_s-\tau_r}{T}\frac{\partial}{\partial \overline{gd}_f}\left(c_0(1-\overline{gd}_f)p_0 + c_1(1-\overline{gd}_d)p_1\right) \\ 0 &= -c_0p_0 + c_1\frac{\partial}{\partial \overline{gd}_f}(1-\overline{gd}_d)p_1 \\ \frac{\partial}{\partial \overline{gd}_f}(1-\overline{gd}_d) &= \frac{c_0p_0}{c_1p_1} \\ \frac{\partial}{\partial \overline{gd}_f}(1-\overline{gd}_d) &= \frac{c_0}{c_1} \end{aligned} \quad (D.1)$$

where, $p_0 = p_1 = 0.5$ and $c_1 = log_2\left(1 + \frac{SNR_{CR-IoT}}{1+SNR_{PU}}\right)$, which consists of the SNR_{PU} in the PU transmitter and the CR-IoT user receiver link. Therefore, the appearance of the PU signal can reduce the achievable throughput of the CR-IoT user transmitter and the CR-IoT user receiver link. □

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